

Important Diagrams for Board Exam & NCERT Page Numbers

Class 11 Biology — Chapter 4: Animal Kingdom

This reference guide catalogs key schematic diagrams, classification charts, and representative organism diagrams from **Chapter 4: Animal Kingdom (NCERT Class 11 Biology)** along with their exact **Figure Numbers, NCERT Page Numbers, and Exam Priority**.

Master List of Important Diagrams

Fig / Table No.	Diagram / Topic Name	NCERT Page	Exam Priority	Key Labels / High-Yield Concepts
Figure 4.16	Chordata Characteristics (Fundamental Plan)	Page 46	★★★ Very High	Nerve cord, Notochord, Gill slits (Pharyngeal gill slits), Post-anal part (tail).
Figure 4.3	Diagrammatic Sectional View of Body Cavity (Coelom)	Page 39	★★★ Very High	(a) Coelomate (True coelom lined by mesoderm), (b) Pseudocoelomate (Scattered mesoderm pouches), (c) Acoelomate (Solid body without cavity).
Figure 4.2	Showing Germinal Layers	Page 38	★★★ Very High	(a) Diploblastic (Ectoderm, Mesoglea, Endoderm), (b) Triploblastic (Ectoderm, Mesoderm, Endoderm).
Figure 4.15	Balanoglossus (Phylum Hemichordata)	Page 45	★★★ Very High	Proboscis, Collar, Trunk.
Figure 4.7	Diagrammatic View of Cnidoblast (Cnidocyte)	Page 41	★★☆ High	Stinging capsule / Nematocyst, operculum, thread tube, anchoring structure.
Figure 4.1	Body Symmetry	Page 38	★★☆ High	(a) Radial symmetry (multiple central planes), (b) Bilateral symmetry (single plane dividing into left and right halves).
Figure 4.6	Body Forms of Coelenterata	Page 41	★★☆ High	(a) Medusa (Umbrella-shaped, Aurelia), (b) Polyp (Cylindrical sessile, Adamsia/Hydra).
Figure 4.10	Sexual Dimorphism in Aschelminthes (Ascaris)	Page 43	★★☆ High	Male (curved posterior tail with cloacal aperture) vs Female (longer, straight body).
Figure 4.18	Petromyzon (Class Cyclostomata)	Page 47	★☆☆ Moderate	Jawless circular sucking mouth, 6–15 pairs of gill slits, lack of paired fins and scales.
Flowchart (Sec 4.2.11)	Classification of Subphylum Vertebrata	Page 47	★★★ Very High	Agnatha (Cyclostomata) vs Gnathostomata (Pisces: Chondrichthyes, Osteichthyes; Tetrapoda: Amphibia, Reptilia, Aves, Mammalia).
Table 4.1 & 4.2	Comparative Tables (Chordates vs Non-chordates & Salient Features)	Page 46, 51	★★★ Very High	Levels of organisation, symmetry, coelom, segmentation, circulatory & respiratory systems across all animal phyla.



High-Yield Board Exam Drawing Tips:

- **Mandatory 3-Mark / 5-Mark Drawing:** *Chordata Characteristics (Fig 4.16, Page 46)* is the most frequently asked schematic drawing. Make sure the relative dorsal-ventral positions of the nerve cord and notochord are accurate.
- **Coelom Types (Fig 4.3, Page 39):** Clear representation of mesoderm lining is critical for scoring full marks in Section A/B conceptual questions.
- **Balanoglossus (Fig 4.15, Page 45):** Always ensure the three clear divisions—Proboscis, Collar, and Trunk—are cleanly delineated and labelled.